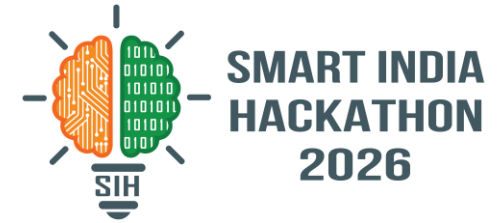


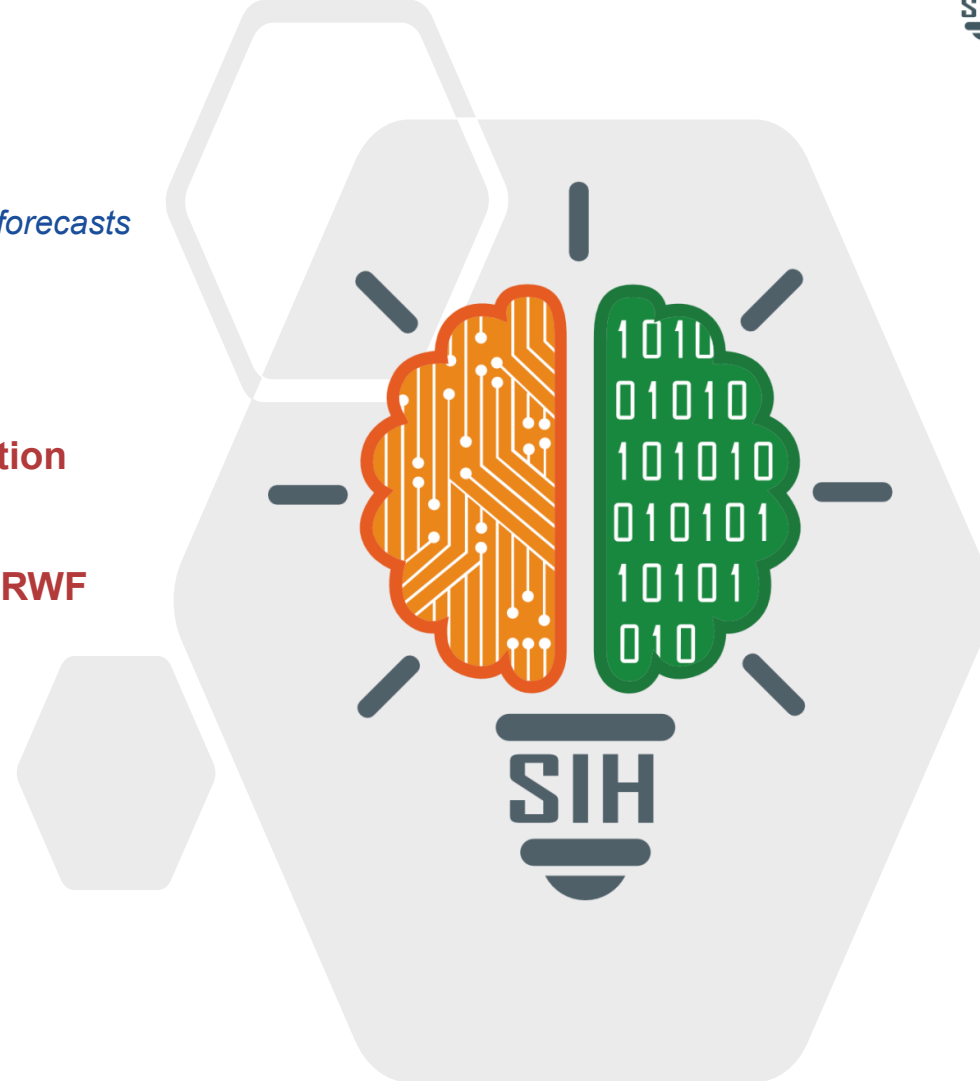
SMART INDIA HACKATHON 2026



विश्वास · VISHWAS

AI forecast-bust detection & trust console for NCMRWF's medium-range forecasts

- Problem Statement ID: **26079**
- Problem Statement Title: **AI-Based Forecast Bust Detection for Medium-Range Weather Forecasts**
- Organisation: **Ministry of Earth Sciences (MoES) – NCMRWF**
- Theme: **Disaster Management**
- PS Category: **Software**
- Team ID: **—**
- Team Name: **PowerPuff Squad**



330

judgements per cycle, ≈ 1 s

23 yrs

of NCMRWF forecast errors

0.02

calibration error on unseen years

90%

of busts flagged at Yellow+

Proposed solution: we rate how wrong NCMRWF's issued forecast will be, before the outcome is known.

! TODAY'S GAP

NCMRWF's duty forecaster has the forecast and the ensemble spread, but no calibrated answer to:

"How likely is *this* forecast, for *this* region and lead day, to be badly wrong?"

Busts cluster in rapidly evolving systems:

Monsoon depressions

Heavy rainfall

Western disturbances

Cyclones

Heat waves

Break / active monsoon

The cost lands downstream: disaster managers, farmers, reservoir, power and transport planners act on a forecast nobody has rated.

WHAT VISHWAS RETURNS, PER CYCLE

330 judgements = 33 subdivisions × Day 1–10, scored in ≈1 s:



Calibrated bust probability

a stated 40% really busts ≈40% of the time



Forecast confidence map, Day 1–10

100% – bust probability, animated per lead day



IMD colour-coded level + action

Green / Yellow / Orange / Red, each with an SOP step



Big-miss watch

separate head for rare, costly large-amount errors



Top-3 meteorological reasons

exact TreeSHAP, written as sentences with real numbers



Closest historical case

the most similar past forecast and how it verified



Model self-confidence

the AI's own precedent, on a separate violet scale

💡 INNOVATION & UNIQUENESS

1

Rates the forecast, not the weather

a trust layer on what NCMRWF already issues; no NWP output changes

2

Learned from NCMRWF's own history

23 years of NCMRWF forecasts vs IMD rainfall

3

Probabilities you can write an SOP on

coastal / inland isotonic calibration, ECE 0.02

4

Two-headed risk

big-miss head lifts large-miss detection 41% → 75%

5

Model Trust: an honest 2nd axis

the AI flags its own low-precedent cases

6

Humans close the loop

approved outcomes recalibrate; IMD standards, bilingual

PROOF, OUT OF SAMPLE

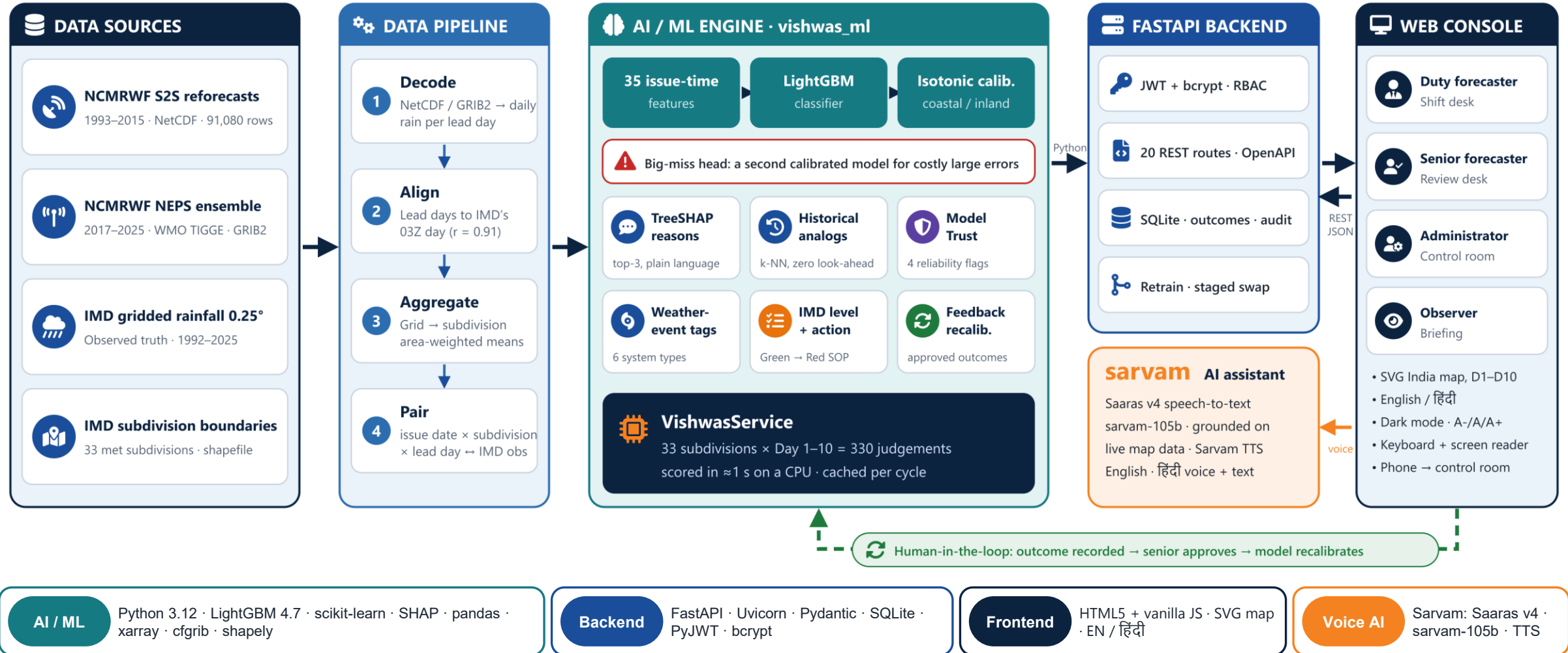
Chennai floods, cycle of 1 Dec 2015: Tamil Nadu Day 4 rated 40%, Orange "Alert". It busted. Flagged 4 days ahead.

D1 · Watch

D4 · 40% Alert

Busted ✓

Architecture: CPU-only, three tiers: Python ML engine → FastAPI REST API → accessible web console



Leak-free by design: all 35 inputs are known at 00Z on the issue date; nothing leaks from the future.

1 · WHAT COUNTS AS A BUST

Magnitude bust

$|\text{forecast} - \text{observed}| \geq \max(25 \text{ mm, that subdivision's 95th-percentile error for the lead day})$

Category bust

the rain / no-rain call was wrong at IMD's 2.5 mm rainy-day line

Bust = magnitude OR category. Thresholds fitted on training years only; admins can tune them.

IMD colour-coded warning ladder

Green
0–18%

No action: forecast is reliable

Yellow
18–38%

Cross-check the short-range nowcast

Orange
38–62%

Escalate for manual review

Red
62–100%

Escalate now: treat with strong caution

2 · FROM FORECAST TO EXPLAINED RISK



35 issue-time features

amount & anomaly · distance to 2.5 mm line · lead-day consistency · pressure, wind, moisture z-scores · out-of-fold bust history · season · terrain & coast



LightGBM classifier

gradient-boosted trees, early stopping on a held-out calibration year



Isotonic calibration

fitted separately for coastal and inland subdivisions, clipped 1–99%



Big-miss head

second calibrated model; flag tuned for 15% precision, never alters the main probability



Explain every flag

TreeSHAP factor sentences · SHAP-weighted k-NN analog · 4-flag Model Trust

Tamil Nadu, Day 4 → 40% Orange. Top reason (+53%):
“Forecast of 2.3 mm sits right at the 2.5 mm rain/no-rain line, so a small error flips the call.”

3 · ASK VISHWAS, BY VOICE



Forecaster asks, by voice or text

“तमिलनाडु में दिन 4 पर जोखिम क्यों है?”



Saaras v4 speech-to-text

auto-detects Hindi / English



sarvam-105b, grounded

attached: live bust %, level, SHAP reasons



Sarvam text-to-speech

answer read aloud in the same language

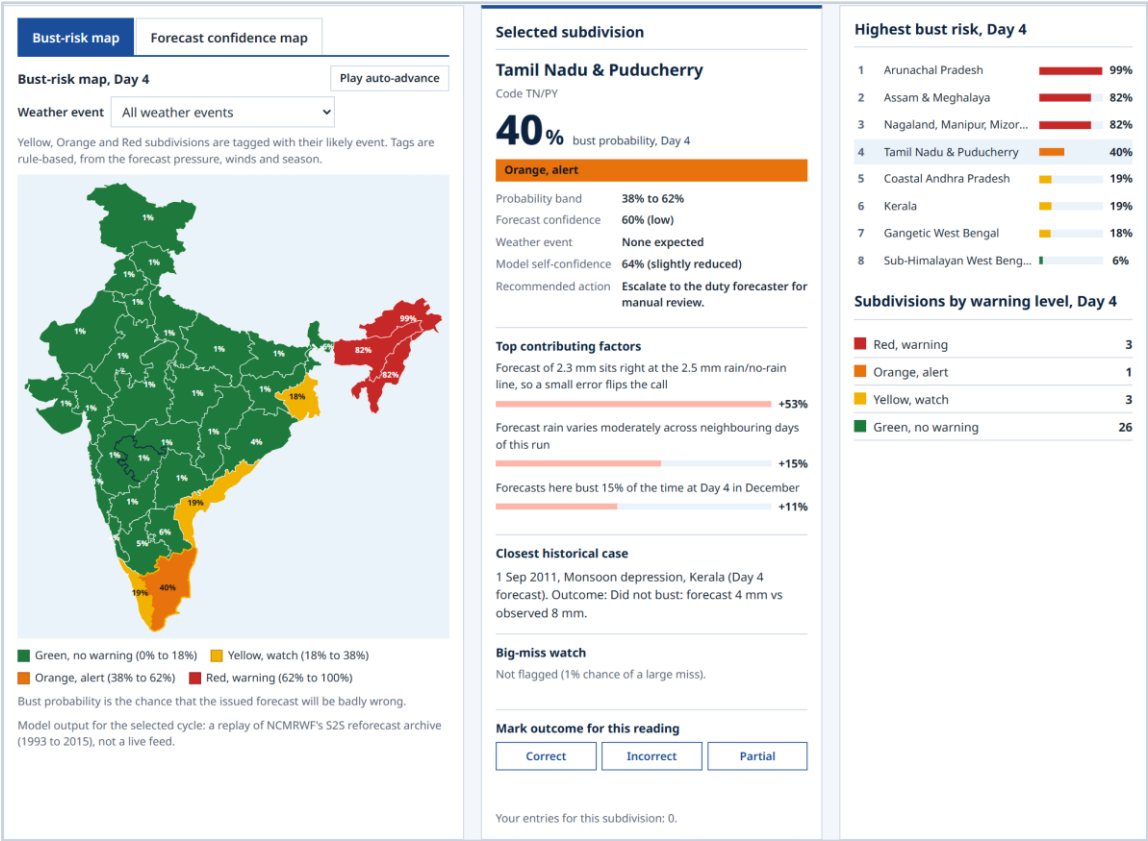


Answer cites the model, never invents

no live data attached → it says so

Powered by Sarvam AI. The API key stays on the server; a failed call degrades to a clean message.

Built, not mocked: real NCMRWF data, tested once on unseen 2013–2015 (11,880 forecasts, 1,976 busts).



Live dashboard, 1 Dec 2015 cycle: bust-risk map, the selected subdivision's reasons and closest case, and the highest-risk list.



Built, tested, cheap to run: uses what NCMRWF and IMD already produce, on one modest server.



Technical feasibility

Built end to end: real data → calibrated model → REST API → role-based console. 88 tests pass; CPU only.



Operational feasibility

Speaks the forecaster's language: IMD subdivisions, colour code and rainfall thresholds; 4 roles mirror the desk.



Data feasibility

Readers for NCMRWF S2S (1993–2015) and NEPS (2017–2025); lead-day alignment verified ($r = 0.77 / 0.91$).



Financial feasibility

100% open-source stack and open data; no per-seat licence; two processes on a single server.



RISKS → HOW VISHWAS HANDLES THEM

Busts are rare (~17%)	PR-AUC as the main metric, calibrated probabilities, dedicated big-miss head
Information leakage	issue-time features only, out-of-fold history, look-ahead-free analogs, automated leakage tests
Over-fitting	year-based splits, separate validation years, test used once, bootstrap CIs
Coastal, hill & plains regimes	coastal / inland calibration, terrain & coast features, per-subdivision thresholds
Forecaster trust in AI	plain-language reasons, historical analogs, Model Trust, calibration proven on unseen years
Archive access & timing	dedicated S2S / TIGGE readers; lead days aligned to IMD's 03Z day
Keeping the model current	approved-outcome recalibration; admin retraining with zero-downtime swap



SUSTAINABLE: LEARNS WITH USE



ROADMAP TO SCALE

- 1 Live NCUM-G / NEPS feed, scored as issued
- 2 District-level trust maps
- 3 Heat waves & cyclones: temperature, wind
- 4 MoES deploy: containers, PostgreSQL, SSO
- 5 Confidence lines in IMD & Agromet bulletins

Impact: a calibrated, explained confidence at issue time puts attention where the forecast is likely to fail.

330

subdivision × lead-day
ratings per forecast cycle

90%

of busts land at Yellow or
above, before the
outcome

75%

of large, costly misses
caught by the big-miss
watch

4 days

of warning on the
Chennai 2015 flood-day
bust

Who benefits



NCMRWF & IMD forecasters

A map read in seconds shows which subdivisions and days need a second look, each with reasons and an action.

Uses: map, reasons, SOP step



Disaster management

Warnings carry a confidence level, so preparedness scales with how reliable the forecast is.

Uses: confidence + big-miss



Farmers & Agromet units

Sowing, irrigation, spraying and harvest advisories can say how far to rely on the forecast.

Uses: confidence per lead day



Reservoir, power & transport

Act on a trustworthy forecast; hedge on an uncertain one.

Uses: Day 1–10 confidence



NWP developers

Per-subdivision SHAP factors show where and why the model errs, by season and lead day.

Uses: SHAP factor record



Social

Better-qualified warnings protect lives and livelihoods; honest confidence builds public trust in official forecasts.



Economic

Fewer costly decisions on unreliable forecasts, fuller use of reliable ones; guidance drives crores of decisions weekly.



Environmental

Better-timed irrigation and reservoir releases save water; trusted heavy-rain guidance cuts flood damage.



Institutional

Every approved outcome builds a verified record of forecast skill by region, season and lead day.

Grounded: built on NCMRWF and IMD data and peer-reviewed forecast-verification science.

DATA SOURCES

- 1 NCMRWF Research Data Server: S2S reforecast 1993–2015. DOI 10.64349/nmrf.rds.s2s.50521 · rds.ncmrwf.gov.in/datasets/s2s-daily
- 2 TIGGE forecasts, origin NCMRWF (NEPS), ECMWF Data Store · ecds.ecmwf.int/datasets/tigge-forecasts
- 3 Pai, D. S. et al. (2014). A 0.25×0.25 degree daily gridded rainfall data set over India. *Mausam*, 65(1), 1–18.
- 4 IMD meteorological subdivision boundaries · github.com/India-Meteorological-Department/Indian_met_zones
- 5 IMD SOP for forecasting & warning: colour code; rainy day ≥ 2.5 mm, heavy rain ≥ 64.5 mm · mausam.imd.gov.in
- 6 NCMRWF operational NCUM-G and NEPS-G prediction systems · ncmrwf.gov.in

METHODS

- 7 Rodwell, M. J. et al. (2013). Characteristics of occasional poor medium-range weather forecasts. *BAMS*, 94(9).
- 8 Ke, G. et al. (2017). LightGBM: a highly efficient gradient boosting decision tree. *NeurIPS* 30.
- 9 Lundberg, S. M. et al. (2020). From local explanations to global understanding with explainable AI for trees. *Nature ML*, 2.
- 10 Zadrozny, B. & Elkan, C. (2002). Transforming classifier scores into accurate probability estimates. *KDD*.
- 11 Saito, T. & Rehmsmeier, M. (2015). The precision-recall plot is more informative than ROC on imbalanced data. *PLoS ONE*, 10(3).
- 12 Wilks, D. S. (2019). *Statistical Methods in the Atmospheric Sciences*, 4th ed. Elsevier.

OUR RESEARCH RIGOUR

- Year-based train / validation / test splits
- Thresholds fitted on training years only
- Test years used exactly once
- Block-bootstrap 95% confidence intervals
- Automated leakage tests
- Cross-checked on NOAA GEFS: +0.042 PR-AUC

Code & handbook: github.com/Tanish-Shitanshu/sih2026-forecast-bust-detection